Event-Triggered Consensus for Multi-Agent Systems with General Linear Dynamics

Project 3, *Analysis and Control of Multi-Robot Systems*
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Abstract

We study the consensus problem for a network of N agents with general linear dynamics $\dot{x}_i = Ax_i + Bu_i$, $y_i = Cx_i$, under a *limited-communication* constraint: each agent broadcasts its state to its neighbours only when strictly necessary, i.e. when a local error exceeds a threshold (*event-triggered control*). The goal is to reduce communication and computation with respect to continuous control while preserving convergence to consensus. The coupling gain is designed through a Riccati inequality (Isidori, Ch. 5.4–5.5); a *model-based* protocol with the decentralized triggering rule of Garcia, Cao and Casbeer (2014) is adopted, including the variant that excludes Zeno behaviour, and the event-triggered scheme is compared in simulation against continuous control. The implementation is validated on the single-integrator case (Dimarogonas et al., 2012), and the communication advantage of the dynamic model over the zero-order hold is reproduced. The communication/convergence trade-off curve obtained by sweeping the threshold parameter σ is the central comparative result. A dedicated analysis separates the two cost axes, *communication* and *computation*, and shows that they do not always move together.

1 Introduction and problem formulation

Objective. In classical consensus every agent updates its control *continuously*, using the instantaneous information of its neighbours; this requires continuous communication over the network. In many applications (embedded microprocessors, limited bandwidth, energy-constrained robots) this is not feasible. The event-triggered idea is to keep a broadcast state value "frozen" and to recompute/rebroadcast *only when* the agent has drifted far enough from that value. The project asks to (i) design an event-triggered consensus controller for general linear dynamics, and (ii) compare it with the classical continuous controller along three axes: communication, computation and performance.

Model. We consider N identical agents

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \qquad y_i(t) = Cx_i(t), \qquad i = 1, \dots, N, \tag{1}$$

with $x_i \in \mathbb{R}^n$, $u_i \in \mathbb{R}^m$, $y_i \in \mathbb{R}^p$ and fixed A, B, C. We seek *output consensus*: there exists $y_{\text{cons}}(t)$ such that $y_i(t) - y_{\text{cons}}(t) \to 0$ for every i. The network is described by an undirected connected graph.

Triggering rule (statement). The basic form of the event condition is

$$\|e_i(t)\| > \sigma \, \|x_i(t)\|, \qquad e_i(t) = x_i(t_{i,k}) - x_i(t), \tag{2}$$

where $t_{i,k}$ is the last trigger instant of agent i and $\sigma > 0$ tunes the communication/performance trade-off. As discussed in §4, this rule is adequate for the single-integrator case but must be extended for general linear dynamics.

2 Background: graph theory and continuous consensus

2.1 Laplacian and connectivity

For an undirected graph with adjacency matrix $\mathscr{A} = [a_{ij}]$ ($a_{ij} = 1$ if i, j are adjacent) and degrees $d_i = \sum_j a_{ij}$, the *Laplacian* is $L = D - \mathscr{A}$, with $D = \mathrm{diag}(d_i)$ and

$$(Lx)_i = \sum_{j \in \mathscr{N}_i} a_{ij}(x_i - x_j).$$

Two facts drive everything that follows. First, L is symmetric, so it is orthogonally diagonalizable, $L = T\Lambda T^\top$ with $T^\top T = I$ and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$; its eigenvalues are real. Second, for any $x = [x_1^\top, \ldots, x_N^\top]^\top$ the associated quadratic form is the *disagreement energy*

$$x^\top (L \otimes I_n) x = \tfrac{1}{2} \sum_i \sum_{j \in \mathscr{N}_i} a_{ij} \left\| x_i - x_j \right\|^2 \ \geq \ 0, \tag{3}$$

which is zero if and only if $x_i = x_j$ for all connected pairs. Hence $L \succeq 0$, and on a connected graph the null space is exactly the *consensus subspace* $\ker L = \mathrm{span}\{\mathbf{1}\}$, so $L\mathbf{1} = 0$ and the eigenvalues order as $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$. The multiplicity of the zero eigenvalue equals the number of connected components; therefore the graph is connected $\iff \lambda_2 > 0$ (the *Fiedler value*, or algebraic connectivity). A larger λ_2 means a "better connected" network and, as shown below, faster consensus. Crucially, λ_2 is the only *global* quantity the design needs: it fixes the minimum coupling gain $c \geq 1/\lambda_2$, and it is computed once, off-line; the run-time protocol uses only local sums $(Lx)_i$.

2.2 Continuous protocol and gain design (Riccati)

With continuous relative information the control is

$$u_i = cF \sum_{j \in \mathscr{N}_i} (x_i - x_j), \qquad F = -B^\top P, \qquad c \geq 1/\lambda_2, \tag{4}$$

where $P \succ 0$ solves the Riccati inequality (with $\theta > 0$)

$$PA + A^\top P - 2PBB^\top P + 2\theta P < 0. \tag{5}$$

This is the observer-free design of Isidori (Ch. 5.5). In stacked (Kronecker) form the closed loop reads

$$\dot{x} = \big[I_N \otimes A - c\,(L \otimes BB^\top P)\big] x. \tag{6}$$

Decoupling by eigenvalue. Because $L = T\Lambda T^\top$ is symmetric, the change of coordinates $\xi = (T^\top \otimes I_n)x$ block-diagonalizes (6): using $(T^\top \otimes I)(L \otimes BB^\top P)(T \otimes I) = \Lambda \otimes BB^\top P$, the Nn-dimensional system splits into N *independent* n-dimensional modes

$$\dot{\xi}_i = \big(A - c\,\lambda_i BB^\top P\big)\xi_i, \qquad i = 1, \ldots, N. \tag{7}$$

These modes carry a clean physical meaning:

- **Consensus mode** ($i = 1$, $\lambda_1 = 0$): $\dot{\xi}_1 = A\xi_1$. This is the component along $\mathbf{1}$; it is *not* required to be stable, since it *is* the common trajectory that all agents follow. Its free evolution $\xi_1(t) = e^{At}\xi_1(0)$ is exactly the reason the consensus value is $Ce^{At}x^\circ$ rather than a constant.
- **Disagreement modes** ($i \geq 2$, $\lambda_i > 0$): these must be Hurwitz for the inter-agent differences to vanish, i.e. for consensus to occur.

Why the Riccati gain works. Fix a disagreement mode $i \geq 2$ and take the Lyapunov function $V_i = \xi_i^\top P \xi_i > 0$. Along (7),

$$\dot{V}_i = \xi_i^\top \left[PA + A^\top P - 2c\lambda_i PBB^\top P\right] \xi_i. \tag{8}$$

Since $c\lambda_i \geq c\lambda_2 \geq 1$ and $PBB^\top P \succeq 0$, we have $-2c\lambda_i PBB^\top P \preceq -2PBB^\top P$, so by the Riccati inequality (5)

$$\dot{V}_i \leq \xi_i^\top \left(PA + A^\top P - 2PBB^\top P\right) \xi_i < -2\theta\, \xi_i^\top P \xi_i = -2\theta V_i. \tag{9}$$

Thus every disagreement mode decays at least as fast as $e^{-\theta t}$, and $A - c\lambda_i BB^\top P$ is Hurwitz. The single scalar condition $c \geq 1/\lambda_2$ makes this hold *simultaneously* for all $i \geq 2$ (the binding case is the smallest positive eigenvalue λ_2). The reduction of a network stability problem to $N-1$ decoupled Hurwitz conditions indexed by the Laplacian eigenvalues is the structural backbone of the whole design, and θ sets a guaranteed convergence rate.

Remark 1. For general A the consensus value is *not* a constant, nor the plain average: the states converge along a free trajectory $y_{\text{cons}}(t) = Ce^{At}x^\circ$, with x° a linear function of *all* the initial conditions. The initial average is recovered only in the single-integrator case ($A = 0$) on a balanced graph.

3 Model-based event-triggered protocol

3.1 Decoupled models and error

Instead of holding the last received value constant (zero-order hold, ZOH), each agent propagates a *model* of the decoupled neighbour dynamics:

$$\dot{y}_i(t) = A\, y_i(t), \quad t \in [t_{i,k}, t_{i,k+1}), \qquad y_i(t_{i,k}) = x_i(t_{i,k}). \tag{10}$$

When agent i triggers, it broadcasts $x_i(t_{i,k})$ to its neighbours and everyone resets $y_i \leftarrow x_i$; since the models share A and the same resets, the copies of y_i held by all agents coincide. We define the model error and the local disagreement signal (both locally measurable):

$$e_i = y_i - x_i, \qquad z_i = \sum_{j \in \mathcal{N}_i} (y_i - y_j). \tag{11}$$

The event-triggered control uses the models:

$$u_i = c_1 F z_i, \qquad c_1 = c + c_2, \quad c_2 > 0. \tag{12}$$

For $A = 0$ the model (10) degenerates into a ZOH, recovering the single-integrator protocol of Dimarogonas et al.

Why a dynamic model, not a hold: the error dynamics. The decisive difference between the two schemes is visible in the dynamics of the model error $e_i = y_i - x_i$. Between two events, with the dynamic model (10),

$$\dot{e}_i = \dot{y}_i - \dot{x}_i = Ay_i - (Ax_i + Bu_i) = A\, e_i - Bu_i. \tag{13}$$

The error is driven *only* by the input term Bu_i, and $u_i = c_1 F z_i \to 0$ as consensus is approached; hence e_i stays small and events become rare. Holding the last value instead ($\dot{y}_i = 0$, i.e. assuming $A = 0$) gives

$$\dot{e}_i = -\dot{x}_i = -Ax_i - Bu_i, \tag{14}$$

now driven by the whole free term Ax_i, which is large (and divergent when A is unstable) even far from consensus. The threshold is then crossed almost immediately, so the ZOH triggers continuously and may even exhibit Zeno behaviour. This is Garcia's key observation: *holding the last value is the same as assuming* $A = 0$, which is exact only for the single integrator. Replacing the held value with the decoupled model $\dot{y}_i = Ay_i$ is precisely the fix that makes event-triggering effective for general (even unstable) dynamics.

3.2 Decentralized triggering rule and convergence

Let $M = PBB^\top P$. The event condition for agent i is

$$\psi_i > \rho\, z_i^\top \Sigma_i z_i, \tag{15}$$

with $0 < \rho < 1$ (the σ parameter) and, setting $d_i = |\mathcal{N}_i|$ and $b_i > 0$,

$$\psi_i = 2(c_2 - c)d_i z_i^\top M e_i + \Big[2cd_i^2(1+b_i) + \tfrac{c_2-c}{b_i} d_i + cd_i(N-1)\big(b_i + \tfrac{3}{b_i}\big)\Big] e_i^\top M e_i, \tag{16}$$

$$\Sigma_i = \big(2c_2 - b_i d_i(c_2 - c)\big)\, PBB^\top P. \tag{17}$$

All quantities in (15) to (17) are *local*, so the rule is fully decentralized: no agent needs global information beyond the design constant λ_2.

Where the rule comes from. The structure of (15) is dictated by the network Lyapunov function $V = x^\top(L \otimes P)x$ of §3.2. Differentiating V along the model-based closed loop and substituting $x_i = y_i - e_i$ produces two kinds of terms: a *nominal* part $x^\top \bar{L} x \preceq 0$ (the same one that gives continuous consensus) plus *error-induced* terms that depend on the model errors e_i. Collecting the latter, ψ_i in (16) is exactly the sum of the degradation caused by e_i: a bilinear part $2(c_2-c)d_i z_i^\top M e_i$ (a cross term in z_i and e_i) and a quadratic part $(\cdots)\, e_i^\top M e_i$; the coefficients follow from Young's inequality with free parameters $b_i > 0$, which is why b_i appears in both ψ_i and Σ_i. The quadratic form $z_i^\top \Sigma_i z_i$ in (17) is the *margin* that the nominal decrease makes available. Triggering when $\psi_i > \rho\, z_i^\top \Sigma_i z_i$ keeps the damage below a fraction $\rho < 1$ of the margin, so a net negative $\dot{V}$ is preserved. At an event the model is reset, $e_i \to 0$, hence $\psi_i \to 0$: the inequality is satisfied with room to spare, and a new inter-event interval begins.

Theorem 1 (Asymptotic consensus)**.** *If (A,B) is controllable, the graph is undirected and connected, $F = -B^\top P$, $c \geq 1/\lambda_2$ and $c_2 > 0$, then the agents (1) with control (12) reach asymptotic consensus whenever events are generated according to (15).*

Proof sketch. With $\hat{L} = L \otimes P$ take $V = x^\top \hat{L} x \geq 0$. Along trajectories one gets $\dot{V} = x^\top \bar{L} x + \sum_i \big[(b_i d_i(c_2 - c) - 2c_2) z_i^\top M z_i + \psi_i\big]$, with $\bar{L} = \hat{L} A_c + A_c^\top \hat{L} \preceq 0$ (Theorem 1 of [1]: $-\bar{L}$ has the same n null eigenvalues as $\hat{L}$, the others negative). At an event $e_i \to 0 \Rightarrow \psi_i \to 0$; enforcing (15) gives $\psi_i \leq \rho z_i^\top \Sigma_i z_i$, hence $\dot{V} \leq x^\top \bar{L} x + \sum_i (\rho - 1) z_i^\top \Sigma_i z_i \leq x^\top \bar{L} x \leq 0$, from which consensus follows. □

3.3 Zeno behaviour and its exclusion

Rule (15) guarantees convergence but not strictly positive inter-event times. The reason is structural: $\Sigma_i \propto PBB^\top P$ has rank at most $m = \operatorname{rank} B$, so when B is thin (here a single input column) Σ_i is only positive *semidefinite*. Hence $z_i^\top \Sigma_i z_i$ can vanish for $z_i \neq 0$; the threshold then collapses to zero and any $\psi_i > 0$ would trigger instantly, producing an accumulation of events in finite time (*Zeno*). The remedy is to raise the threshold by a strictly positive constant $\phi > 0$:

$$\psi_i > \rho\, z_i^\top \Sigma_i z_i + \phi. \tag{18}$$

Theorem 2 (Bounded consensus, anti-Zeno)**.** *Under the assumptions of Theorem 1 and with rule (18), the agents reach* bounded consensus

$$\lim_{t\to\infty} \big\|x_i(t) - x_j(t)\big\|^2 \leq \frac{N\phi}{\eta\, \lambda_{\min}(P)}, \qquad \eta = \frac{\lambda_{\min \neq 0}(-\bar{L})}{\lambda_{\max}(\hat{L})} > 0,$$

and the inter-event times are lower-bounded by some $\tau_i > 0$ (no Zeno).

Proof idea (no Zeno). Immediately after an event $e_i(t_{i,k}) = 0$. Between events e_i obeys (13), $\dot{e}_i = Ae_i - Bu_i$, so $\|e_i\|$ grows at a bounded rate and ψ_i, being a quadratic form in e_i that vanishes at $e_i = 0$, grows continuously from 0. To violate (18) it must first exceed the strictly positive floor ϕ; this requires a finite growth of $\|e_i\|$ and therefore a strictly positive time $\tau_i > 0$. A uniform lower bound on the inter-event interval follows, exactly as in the single-integrator case where it is explicit, $\tau = \sigma/(\|L\|\,(1+\sigma))$.

Proof idea (the bound). With $V = x^\top(L\otimes P)x$, enforcing (18) leaves an extra $+N\phi$ compared with Theorem 1, giving $\dot V \le x^\top \bar L x + N\phi \le -\eta\, V + N\phi$, where $-\bar L \succeq \eta\,(L\otimes P)$ on the disagreement subspace defines $\eta > 0$. Thus V decays until $V \le N\phi/\eta$ and stays there; converting the P-weighted bound into the Euclidean disagreement through $\lambda_{\min}(P)$ yields the stated inequality. □

The constant ϕ therefore offers a further explicit trade-off: large ϕ reduces communication but enlarges the consensus radius $\propto \sqrt{\phi}$; $\phi \to 0$ recovers the asymptotic behaviour of Theorem 1 (at the price of losing the guaranteed dwell time). This is the mechanism behind the residual plateau observed on the unstable agents in §5.2.

4 The σ-rule and the single-integrator case

Continuous first-order agreement. The single integrator $\dot x_i = u_i$ with $u_i = -\sum_{j\in\mathcal{N}_i}(x_i - x_j)$ gives the closed loop $\dot x = -Lx$. On an undirected connected graph the states converge to the *initial average* $\bar x = \frac{1}{N}\sum_k x_k(0)$. The proof is short and worth stating, because it explains why the single integrator is special. The average $\bar x(t) = \frac{1}{N}\mathbf{1}^\top x$ is invariant:

$$\tfrac{d}{dt}\left(\mathbf{1}^\top x\right) = \mathbf{1}^\top \dot x = -\mathbf{1}^\top L x = -(L\mathbf{1})^\top x = 0, \tag{19}$$

using $L = L^\top$ and $L\mathbf{1} = 0$; meanwhile the disagreement modes ($\lambda_i > 0$) decay, so $x_i \to \bar x(0)$.

Remark 2. Invariance (19) relies on $\mathbf{1}^\top L = 0$, i.e. on an *undirected or balanced* graph. On a general unbalanced digraph the states still agree but on a *weighted* average given by the left zero eigenvector of L, not on the plain mean. This is a common conceptual pitfall.

Event-triggered version (Dimarogonas). Sampling the state and holding the control gives

$$u(t) = -Lx(t_k), \qquad t \in [t_k, t_{k+1}), \qquad e(t) = x(t_k) - x(t), \tag{20}$$

so that $\dot x = -L(x+e) = -Lx - Le$. With the ISS Lyapunov function $V = \frac{1}{2}x^\top L x$ (the disagreement energy, $V = 0 \iff$ consensus on a connected graph),

$$\dot V = -\|Lx\|^2 - (Lx)^\top L e \le -\|Lx\|^2 + \|L\|\,\|e\|\,\|Lx\|\,. \tag{21}$$

Enforcing the relative threshold $\|e\| \le \sigma\,\|Lx\|\,/\,\|L\|$ yields

$$\dot V \le (\sigma - 1)\,\|Lx\|^2\,, \tag{22}$$

which is negative for $0 < \sigma < 1$: this is exactly the input-to-state-stability of $\dot x = -Lx$ with respect to the measurement error e, the property the event-triggered design exploits. The event fires when $\|e\| = \sigma\,\|Lx\|\,/\,\|L\|$, at which point e is reset to zero; a strictly positive inter-event time

$$\tau = \frac{\sigma}{\|L\|\,(1+\sigma)} > 0 \tag{23}$$

excludes Zeno. This is the base case of (10) to (15) with $A = 0$: the model is a ZOH, and (14) reduces to $\dot e = -Bu$ with no divergent free term.

Why the σ-rule does not extend. For general linear dynamics $\|x_i\|$ does not decay (it may even grow for unstable A), so the absolute threshold $\sigma\,\|x_i\|$ does not contract and, as (14) shows, the ZOH error is driven by Ax_i: communication is not reduced over time and Zeno may be unavoidable. This is precisely why the project adopts the model-based protocol with rule (15) to (18), which depends on the *relative* model signal z_i rather than on the absolute state norm.

5 Implementation and results

Numerical setup. The implementation (MATLAB) integrates both the true dynamics and the models with fixed-step RK4 at $\Delta t = 10^{-3}$, checking the event condition at every step; the seed `rng(0)` is used throughout for reproducibility. The metrics computed are: number of triggers per agent, convergence time (first instant at which the disagreement norm falls below 10^{-2} of its initial value), the inter-event time distribution, and the sweep over σ. Three cases are studied, followed by an output-only extension.

5.1 Experiment 1: oscillator (marginally stable LTI)

Five agents with $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)$, $C = [1\ 0]$ (output = position), on a 5-cycle with one chord, so that $\lambda_2 = 1.382$, $c = 1/\lambda_2 = 0.724$ and $c_1 = 1.5/\lambda_2 = 1.085$. Four figures characterise this experiment: the convergence, the synchronised trajectories, the triggering pattern, and the trade-off curve.

Convergence (Fig. 1). The figure plots the disagreement norm $\|\cdot\|_F$ on a *logarithmic* vertical axis against time, for the continuous law (solid) and the event-triggered law (dashed). Both curves are essentially straight lines: the disagreement decays *exponentially*, exactly as the mode-wise bound (9) predicts. The two curves are almost superimposed, the event-triggered one lying slightly above; it reaches the 1% tolerance at $t_{\text{conv}} = 4.87$ s against 4.55 s for the continuous baseline, a penalty of about 7%. The small, roughly constant vertical gap between the two lines is the persistent price of the model error e_i that accumulates between consecutive events, consistent with (13). The reading is therefore: near-identical convergence, obtained with 918 transmissions instead of the continuous exchange of the baseline.

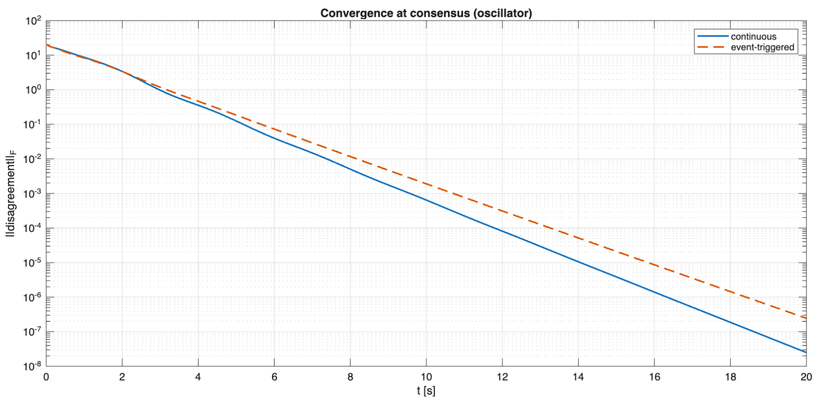

Figure 1: Disagreement norm (log scale) versus time: continuous (solid) and event-triggered (dashed). Both decay exponentially; the event-triggered curve sits slightly above, reaching the tolerance at 4.87 s against 4.55 s. The gap is the cost of the model error e_i between events.

Synchronised states and outputs (Fig. 2). The left panel shows the first state component $x_{i,1}(t)$ of the five agents, the right panel their outputs $y_i = Cx_i$. The trajectories start widely spread (from about $+7$ to -9) and, within roughly five seconds, lock onto a single *common oscillation* of amplitude ≈ 4 and period 2π (the natural period of A). Two theoretical points are confirmed here. First, the agents agree on a shared *trajectory*, not on a constant: this is the free response $Ce^{At}x^{\circ}$ of Remark 1, the hallmark of general A. Second, because $C = [1\ 0]$ selects the position, the outputs panel coincides with the state panel, so output consensus is attained. The brief overshoot before locking is the transient of the second-order oscillatory modes.

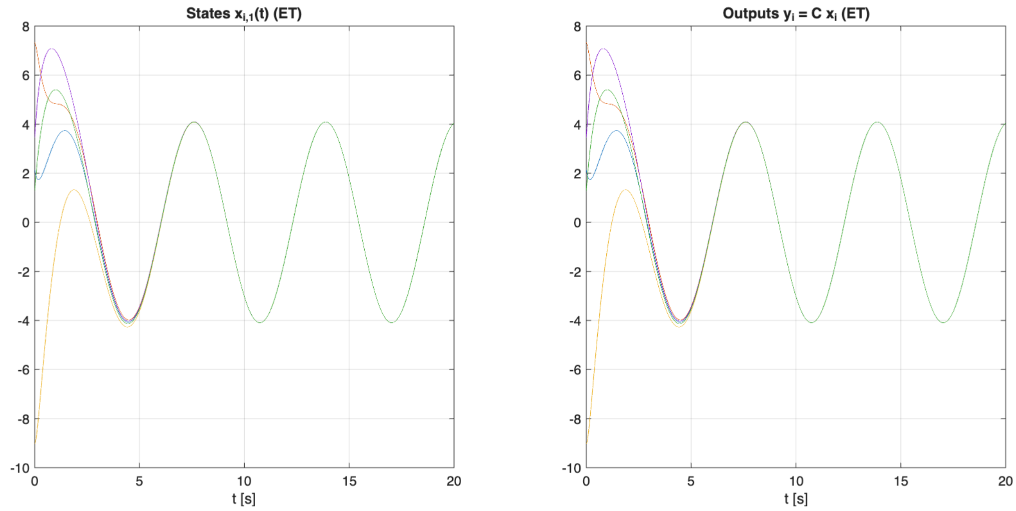

Figure 2: States $x_{i,1}$ (left) and outputs $y_i = Cx_i$ (right) under event-triggered control. The five trajectories start spread out and lock onto a common oscillation of amplitude ≈ 4 within about five seconds; the output panel coincides with the state panel because C selects the position.

Triggering pattern and inter-event times (Fig. 3). The figure has three panels. The left one is the *raster* of transmission instants: each vertical tick is an event, one row per agent, over the whole horizon. Events are dense during the initial transient ($t < 4$ s, when the disagreement is large) and thin out as consensus is approached; crucially they are *asynchronous* across agents, so there is no common clock and the scheme is genuinely decentralised. The centre panel zooms on $[0,3]$ s to make the individual events legible. The right panel is the histogram of inter-event times τ: the mass concentrates around ≈ 0.11 s (the mean) and, above all, lies entirely to the *right* of the red line $\tau_{\min} > 0$. This strictly positive minimum inter-event time is the numerical confirmation of the no-Zeno property, the dwell time guaranteed by Theorem 2. Over the five agents there are 918 transmissions in total (per agent $198, 180, 178, 180, 182$), a nearly uniform split because the agents are identical and almost symmetric on the graph.

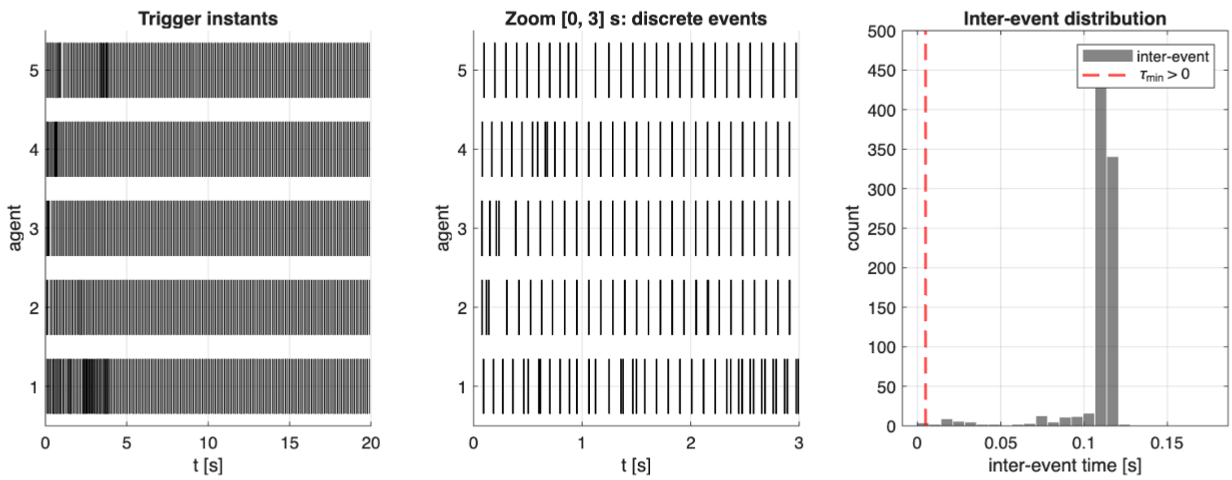

Figure 3: Left: raster of trigger instants per agent over the full horizon (dense in the transient, sparse near consensus, asynchronous across agents). Centre: zoom on $[0,3]$ s. Right: histogram of inter-event times, whose mass sits well to the right of $\tau_{\min} > 0$ (no Zeno).

The σ trade-off (Fig. 4). This is the central comparative result. The threshold parameter ρ (the σ knob) is swept from 0.1 to 0.9 and two quantities are tracked. The blue curve (left axis) is the total number of triggers, which falls monotonically from 1891 to 649 as the threshold is loosened; the orange curve (right axis) is the convergence time, which rises monotonically from 4.68 s to 5.017 s. The two opposite monotone trends make the communication versus performance trade-off explicit and quantitative: a single scalar buys network savings at the price of speed. The trigger curve is *convex*, i.e. there are diminishing returns: the first increments of ρ eliminate many events, later ones progressively fewer. The nominal operating point $\rho = 0.5$ sits mid-curve at ≈ 918 triggers and ≈ 4.87 s, which cross-checks the sweep against the convergence experiment above.

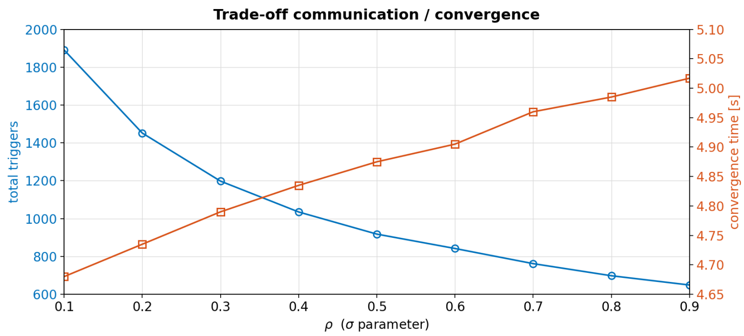

Figure 4: Communication/convergence trade-off versus ρ (the σ parameter): total triggers (blue, left axis, $1891 \to 649$) and convergence time (orange, right axis, $4.68 \to 5.017$ s). The two curves are monotone in opposite directions; the operating point $\rho = 0.5$ gives 918 triggers and 4.87 s.

5.2 Experiment 2: unstable agents, model-based versus ZOH

We reproduce the example of [1]: three third-order agents with unstable dynamics on a path graph (nodes $1,2,3$), under the anti-Zeno rule (18). The triggering rule is kept identical for the two controllers; only the between-event model differs, the dynamic model (10) against a ZOH ($\dot{y}_i = 0$).

Disagreement and communication (Fig. 5). The figure has two panels that together tell the whole story. The left panel is the disagreement on a logarithmic axis: both schemes decay together during the transient, then the model-based curve (solid) settles on a *plateau* at $\approx 2 \times 10^{-2}$, while the ZOH curve (dashed) keeps decreasing to $\approx 10^{-5}$. The right panel counts the transmissions per agent: the model-based bars $(55, 68, 57)$ are barely visible next to the ZOH bars $(6793, 7440, 6723)$. Read together, the panels quantify the decisive advantage of the predictive model: it reaches *bounded* consensus with only 180 transmissions in total, whereas the ZOH triggers 20956 times, that is, essentially continuous communication. The reason is (14): on unstable dynamics the held value drifts almost immediately, the model error explodes, and events pile up; the dynamic model instead follows the free dynamics and communicates only to correct the small input-induced drift.

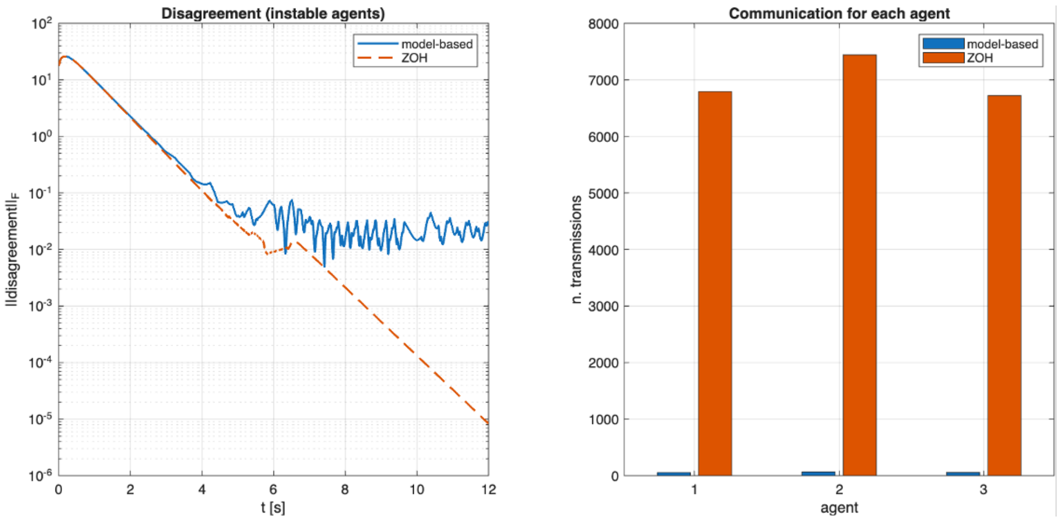

Figure 5: Unstable agents. Left: disagreement on a log axis; the model-based scheme (solid) reaches the bounded-consensus plateau of Theorem 2 at $\approx 2 \times 10^{-2}$, while the ZOH (dashed) decays lower but communicates almost continuously. Right: transmissions per agent; the model-based bars $(55, 68, 57)$ are negligible next to the ZOH bars $(6793, 7440, 6723)$.

Remark 3. In the left panel of Fig. 5 the model-based residual settles slightly *higher* than the ZOH curve. This is not a defect. Under the anti-Zeno rule the model-based scheme deliberately accepts a bounded disagreement (the Theorem 2 ball set by ϕ) in exchange for very few transmissions, whereas the ZOH attains a lower residual only by communicating almost continuously. The plateau height is therefore a design choice, not an error.

5.3 Validation: single-integrator (Dimarogonas)

Five single integrators ($A = 0, B = C = 1$) with rule (15). The states converge *exactly* to the initial average ($\bar{x}(0) = 1.29364$; final states all equal to 1.2936 up to 10^{-9}), confirming the correctness of the implementation on this known case, in agreement with the mean-invariance property (19). The event-triggered scheme uses 451 transmissions in total (see §5.4), against the continuous baseline; the corresponding communication and computation counts are reported and discussed in Fig. 6 below.

5.4 Communication versus computation

The project asks for a comparison not only of performance but also of the *computational* cost. We distinguish two resources: *communication* (number of transmissions on the network) and *computation* (number of control recomputations, i.e. instants at which u_i changes value). In continuous control both are proportional to the number of steps, because the control is updated at every instant. These two axes do *not* always move together, and Fig. 6 together with Table 1 shows exactly when they diverge.

Single-integrator, both axes reduced (Fig. 6). The figure is a grouped bar chart on a logarithmic count axis, blue for communication and orange for computation, for the continuous and the event-triggered controllers. For the continuous law the two bars are equal and maximal (75 000 each): the control is transmitted and recomputed at every step. For the event-triggered law both bars collapse, communication to 451 (about 166$\times$ fewer) and computation to 1430 (about 52$\times$ fewer). One detail is worth reading off the chart: the computation bar (1430) is taller than the communication bar (451). The reason is that a single broadcast by agent i forces a control recomputation not only at i but also at each of its neighbours, whose signal z_i changes; with average degree ≈ 2 this gives about three recomputations per transmission. This is the ideal case $A = 0$, where the held model is exact (§4) and event-triggering wins on both axes at once.

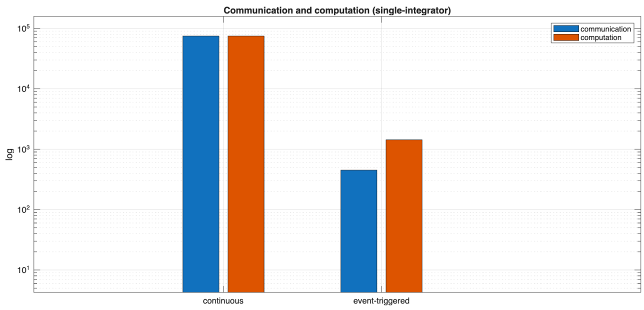

Figure 6: Single-integrator, communication and computation on a logarithmic count axis (blue = communication, orange = computation). Continuous updates both 75 000 times; event-triggering drops communication to 451 and computation to 1430. Computation exceeds communication because each broadcast triggers a recomputation at the sender and at its neighbours.

General dynamics, the two axes separate. For general LTI the model-based control depends on the models, which evolve *continuously*, so u_i changes at every instant even between events. Table 1 compares three strategies on the oscillator. The model-based scheme collapses communication (918 against 100 000) but leaves computation unchanged (100 000). Holding the control constant between events (ZOH) instead reduces computation (71 520), but on oscillatory dynamics the model error grows fast and communication explodes (76 550), so the main advantage is lost.

Table 1: Communication and computation on the oscillator (horizon $T = 100$ s, $\Delta t = 10^{-3}$).

Controller	Communication	Computation
Continuous	100 000	100 000
ET model-based	918	100 000
ET held control (ZOH)	76 550	71 520

This yields a design criterion. The *model-based* scheme is preferable when the bottleneck is the network (bandwidth, transmission energy) and on-board computation is cheap. The *held control* is preferable when the limit is the processor or actuator, but only if events stay sparse (stable dynamics), as in the single integrator. The unstable case of Fig. 5 is the extreme: there the ZOH degenerates into de facto continuous communication.

6 Output consensus with output-only measurement (observer-based)

In the previous cases the full state is transmitted. We now consider the case in which each agent measures *only* the output $y_i = Cx_i$ (with (A,C) observable). The solution combines the observer-based structure of Isidori (Ch. 5.5) with the model-based event-triggered consensus: each agent reconstructs its own state with a local observer and consensus is run on the *estimates*.

Local observer. Each agent runs

$$\dot{\hat{x}}_i = A\hat{x}_i + Bu_i + L_{\text{obs}}\,(y_i - C\hat{x}_i), \tag{24}$$

with L_{obs} chosen so that $A - L_{\text{obs}}C$ is Hurwitz (possible since (A,C) is observable). Subtracting (24) from (1) the input term cancels:

$$\dot{\tilde{x}}_i = (Ax_i + Bu_i) - \big(A\hat{x}_i + Bu_i + L_{\text{obs}}C\tilde{x}_i\big) = (A - L_{\text{obs}}C)\,\tilde{x}_i. \tag{25}$$

The estimation error $\tilde{x}_i = x_i - \hat{x}_i$ therefore obeys an *autonomous* stable equation and decays to zero independently of the consensus dynamics, which is the observer half of the separation argument. For the oscillator with $C = [1\ 0]$ and $L_{\text{obs}} = [\ell_1\ \ell_2]^\top$,

$$A - L_{\text{obs}}C = \begin{pmatrix} -\ell_1 & 1 \\ -1-\ell_2 & 0 \end{pmatrix}, \qquad \det(sI - (A - L_{\text{obs}}C)) = s^2 + \ell_1 s + (1+\ell_2);$$

matching the target $(s+3)(s+4) = s^2 + 7s + 12$ gives $\ell_1 = 7$, $\ell_2 = 11$, i.e. poles $\{-3,-4\}$, deliberately faster than the consensus dynamics so that the estimate is accurate before consensus takes effect.

Event-triggered consensus on the estimates. The estimate $\hat{x}_i$ is transmitted/modelled through χ_i ($\dot{\chi}_i = A\chi_i$, reset $\chi_i \leftarrow \hat{x}_i$ at an event), and the control is

$$u_i = c_1 F z_i, \qquad z_i = \sum_{j \in \mathcal{N}_i} (\chi_i - \chi_j), \tag{26}$$

with the same triggering rule (15) applied to the model error $e_i = \chi_i - \hat{x}_i$. Two error sources are thus separated: the *observation* error $\tilde{x}_i$, handled by L_{obs}, and the *communication* error e_i, handled by the triggering (a separation-principle argument). For $C = I$ the observer becomes trivial ($\hat{x}_i \equiv x_i$) and the full-state protocol is recovered exactly.

Results (Fig. 7). We use the oscillator with $C = [1\ 0]$ (only the position is measured, not the velocity) and observer poles $\{-3,-4\}$. The figure has three panels. The left one shows the outputs y_i: they synchronise onto the common oscillation exactly as in the full-state case, even though only the position is measured. The centre panel shows the observation error $\|x_i - \hat{x}_i\|$ of each agent on a logarithmic axis: it decays as a straight line from order one down to machine precision ($\approx 8.88 \times 10^{-16}$), the numerical signature that $A - L_{\text{obs}}C$ is Hurwitz and the estimates track the true states, as in (25). The right panel shows the state disagreement $\|x - \bar{x}\|_F$, again a straight line on the log axis, decaying to $\approx 2 \times 10^{-7}$. The key point is that the two error panels decay *independently*: the observation error (centre), governed by L_{obs}, and the consensus error (right), governed by the triggering. This is the separation principle in action, and it shows that output consensus is achieved with output-only measurement, at a cost of 1044 triggers (per agent $208, 183, 223, 225, 205$), comparable to the full-state case.

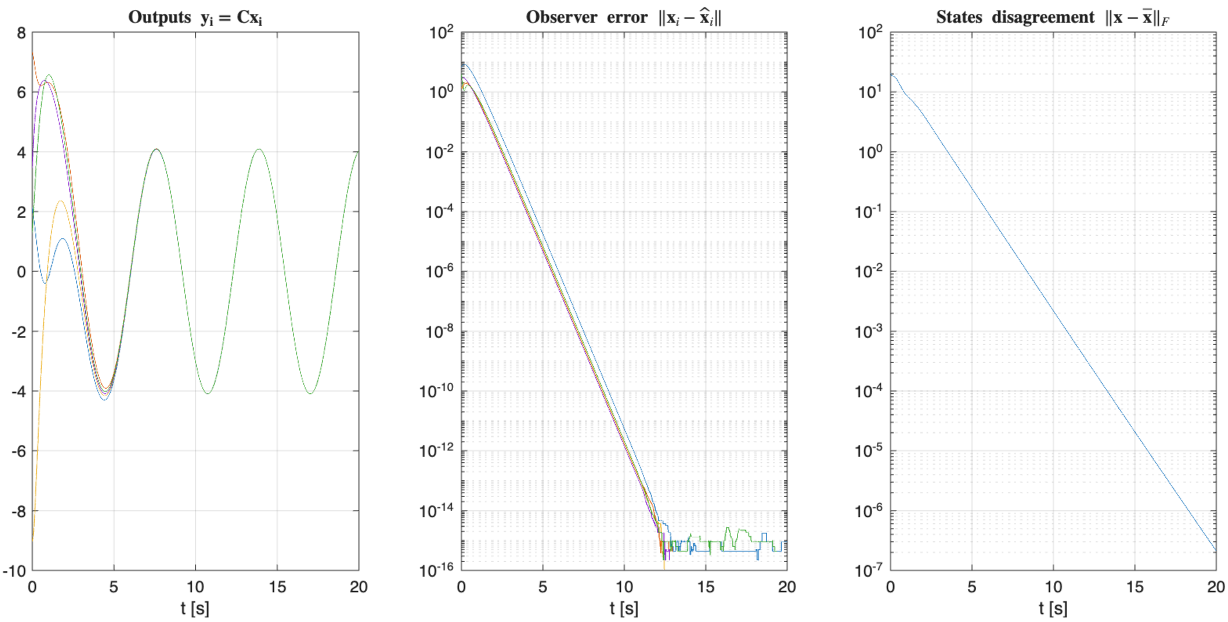

Figure 7: Observer-based output consensus with output-only measurement $y_i = Cx_i$. Left: the outputs synchronize. Centre: the observation error $\|x_i - \hat{x}_i\|$ decays to machine precision. Right: the state disagreement converges ($\sim 2 \times 10^{-7}$).

7 Conclusions

We designed and implemented an event-triggered consensus controller for general linear dynamics and compared it against continuous control. Following a single logical thread (network Laplacian $\to$ eigenvalue decoupling $\to$ Riccati gain $\to$ event-triggering $\to$ extensions), the model-based protocol with a decentralized triggering rule reaches consensus (asymptotic, or bounded with Zeno exclusion) while reducing communication by orders of magnitude; the predictive model is markedly superior to the ZOH on unstable dynamics (180 against 20 956 transmissions). The sweep over σ quantifies the trade-off between communication saving and convergence speed, which is the core of the required comparison. A dedicated analysis of the two cost axes shows an asymmetry that is easy to overlook: for the single integrator event-triggering cuts both communication and computation, whereas for general LTI the model-based scheme cuts communication but not computation, and the held-control variant does the opposite. Finally, the output-only case was solved (§6) by combining a local observer with event-triggered consensus on the estimates, achieving output consensus with the same communication efficiency.

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